

# CS6103 – Introduction to Java

## Spring 2026 – Final Project Proposal

### Project Title:

Peer-to-Peer File Sharing System with Concurrent Chunk Transfers

**Team Members:** Yinkui Yu, Kai-En Huang, Yuxin Zhu

### 1. Overview

This project implements a peer-to-peer (P2P) file sharing system in Java. Peers share and download files directly from each other, coordinated by a lightweight tracker server that maintains metadata (file locations) but does not store file content. Files are transferred in chunks and downloaded concurrently from multiple peers.

### 2. Tech Stack

- **Language:** Java (JDK 17+)
- **Networking:** Socket, ServerSocket
- **Concurrency:** Thread, ExecutorService, synchronization
- **GUI:** JavaFX
- **Database:** SQLite

All networking and concurrency logic will be implemented manually without high-level frameworks.

### 3. Architecture

- **Tracker Server:**  
Maintains file metadata (file → peers), handles registration and search
- **Peer Client:**  
Each peer acts as both client and server:
  - downloads files from others
  - serves file chunks via local server
  - runs GUI + transfer threads

### 4. Core Features

- File sharing and registration
- File search via tracker

- Chunk-based file splitting
- Concurrent multi-threaded downloads
- Direct peer-to-peer transfer
- GUI with progress tracking

## 5. Advanced Concepts

- **Networking:** socket-based peer ↔ peer and peer ↔ tracker communication
- **Multithreading:** parallel chunk downloads and upload handlers
- **Synchronization:** thread-safe tracking of chunk progress and shared state
- **GUI:** interactive desktop interface
- **Database:** persistent metadata and download history

## 6. Workflow

1. Peer registers files with tracker
2. User searches for file
3. Tracker returns available peers
4. Peer downloads chunks concurrently
5. Chunks are reassembled into final file

## 7. Outcome

A working distributed system with:

- multi-peer file transfer
- concurrent downloads
- GUI interface
- database-backed metadata

This project demonstrates networking, multithreading, synchronization, and GUI development beyond a simple CRUD system.